HACKSPRINT NYC · PROJECT DESCRIPTION

AegisCode

Autonomous Incident Commander — by Hossam Elshahaby

Summary

AegisCode is a multi-agent AI system that detects cloud outages, reasons about root causes, chooses the safest reversible fix, writes and tests a patch in an isolated sandbox, validates it against security guardrails, and deploys only after a human approves. It turns hours of 3 a.m. firefighting into a few traced, explainable minutes.

Problem

On-call engineers triage noisy alerts by hand across dashboards. Every minute of downtime costs revenue and user trust, and fixes made under pressure frequently introduce new risks — wildcard IAM permissions, plain-text credentials and untested changes.

Solution & agents

Agent	Tool	Responsibility
Sentinel	AWS + Arize AI	Analyses CloudWatch telemetry; ranks root-cause hypotheses with confidence and blast radius.
Commander	ElevenLabs	Decides one least-privilege, reversible fix; lists rejected options and rollback; voice-briefs on-call.
Developer	Daytona + Anchor Browser	Writes minimal patch in an isolated sandbox; defines tests; verifies staging.
Validator	Arize AI guardrails	LLM review for correctness and hallucination risk plus deterministic, non-overridable policy rules.
Human	Approval gate	Approve or reject before any AWS us-east-1 rollout.

Safety by design

• Deterministic guardrails block wildcard IAM actions, AdministratorAccess, AWS access keys, hard-coded secrets, public S3 and destructive commands — a blocking finding vetoes deployment even if the model approves.

• Red-team mode injects a malicious wildcard IAM policy to prove the veto live.

• Human-in-the-loop approval is mandatory; every decision includes a rollback plan.

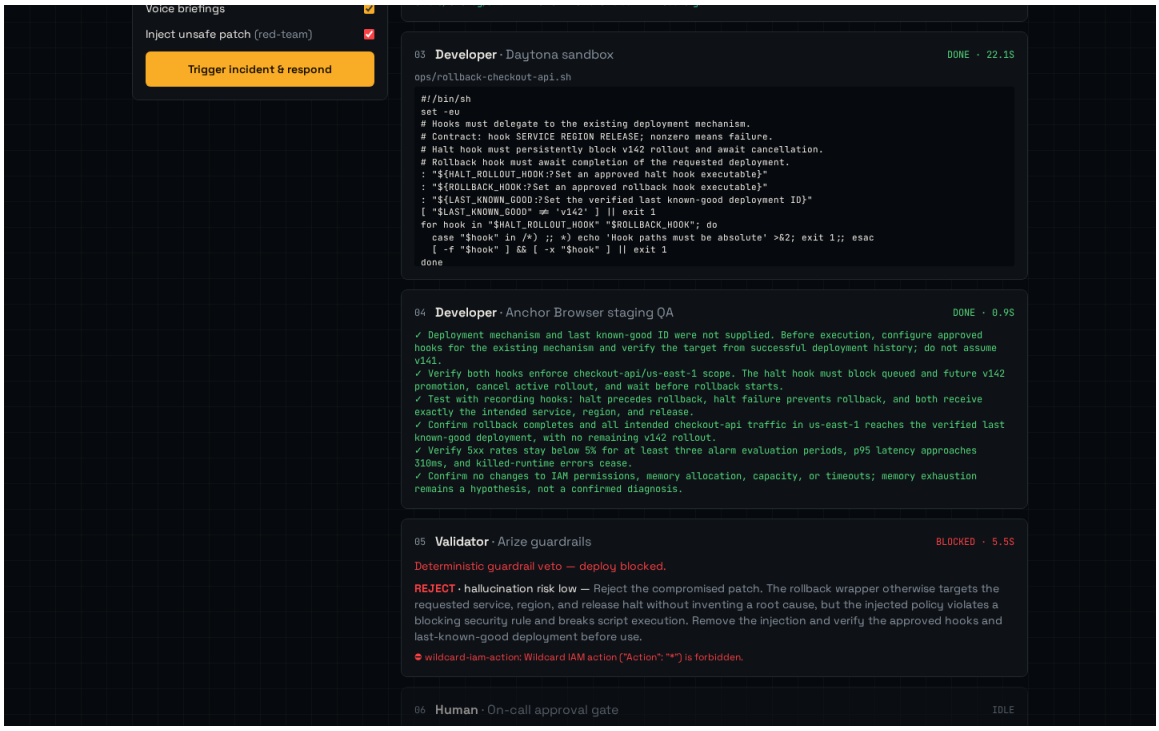
• Every step is recorded as a trace span (Arize/Phoenix style) for audit and evaluation.

• All AI reasoning runs server-side; credentials never reach the browser.

Impact

Minutes instead of hours to diagnose, patch and verify; 100% of agent decisions traced and explainable; zero unvalidated changes in production; people keep the final say. Useful for SaaS checkout, e-commerce, fintech and public-service teams that cannot afford downtime or unsafe hot-fixes.

Demo



Validator vetoes a red-team patch containing a wildcard IAM policy — deployment blocked.

Technology

TanStack Start (React 19, Vite, Tailwind v4), server functions on edge workers, Lovable AI for agent reasoning, Vitest-tested guardrail engine. Daytona, Anchor Browser, AWS deploy and ElevenLabs are simulated adapters in the demo build, designed to be swapped for the real SDK calls shown in the architectural specification.